JOHNSON BOLHAYON

Data Specialist | Spreadsheet Expert | Data Automation Engineer | Web and Mobile Application Developer (AI-assisted)

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PROFILE

Data automation and migration specialist with 9+ years of experience turning repetitive, high-volume data work into automated systems. Started as a spreadsheet expert in advanced Excel and VBA, and now designs, builds and operates full-stack internal web applications with Next.js, TypeScript and Supabase (PostgreSQL), deployed to Google Cloud and integrated with Lightspeed, Xero, Jira, Freshdesk and Confluence through REST APIs. Owns the whole pipeline - extract, clean, validate, migrate - then builds the tools, dashboards and integrations teams rely on afterwards, with every automated output validated cell-for-cell against the report it replaces. Years of hand-written VBA, C# and SQL ETL work predate the AI boom; adopted AI-assisted engineering early and uses Claude Code to deliver production software solo at the pace of a team, every change spec-driven, reviewed and verified. Ships AI into the product itself: an autonomous Gemini agent that investigates and drafts answers to franchise support tickets, and SukliCo POS, an own offline-first mobile point of sale for small Philippine stores. Known for reducing repetitive work by 70-90% and for bridging business workflows with technical solutions.

CORE SKILLS

AI Engineering: Google Gemini API agents - function calling (tool loops), structured output (responseSchema), model fallback chains, cache-aware prompting, per-feature quotas and audit logging; LLM evaluation - replay harnesses with an LLM grader, precision-gated auto-accept; generative media - Gemini image generation, Gemini-TTS and Google Cloud Text-to-Speech; Claude Code - spec-driven, verification-first delivery

Spreadsheets (Expert): Microsoft Excel - advanced formulas, VBA, pivot tables, Power Query, dashboards; Google Sheets

Data and Analytics: Data migration, data cleansing and validation, deduplication, reconciliation, QA testing, Power BI, dashboards

Automation and Scripting: Excel VBA, Google Apps Script, C# (LINQ), ETL pipelines, bulk data imports

SQL and Databases: SQL (MySQL, SQL Server/SSMS, Firebird), Snowflake, PostgreSQL - row-level security (RLS), migration-managed schemas, pgTAP tests; Supabase (Auth, Storage, Edge Functions); SQLite (offline-first, WAL, outbox sync); audit logging

APIs and Integrations: REST API integration, Google Sheets and Drive APIs (service accounts), Lightspeed X-Series API, Xero API, Atlassian REST APIs (Jira, Confluence), Freshdesk API and webhooks, Google Gemini API, Google Play Billing (server-side verification, RTDN), OAuth and scoped API tokens, HMAC request signing, rate-limit handling

Web Development: Next.js (App Router), React, TypeScript, Tailwind CSS, Node.js, NextAuth (Auth.js), Google OAuth, role-based access control (RBAC), Git, Vercel

Mobile: React Native / Expo (SDK 57), Expo Router, Reanimated, Zustand, expo-sqlite offline-first sync, Kotlin native modules, Bluetooth ESC/POS receipt printing, camera barcode and QR scanning, EAS Build

Video and Media Pipelines: Remotion (programmatic video in React), FFmpeg (compositing, camera moves, silence detection), headless Chromium rendering (Playwright), Google Cloud Text-to-Speech

Cloud and DevOps: Google Cloud Platform (GCP) - Cloud Run and Cloud Run Jobs, Cloud Build (CI/CD), Artifact Registry, Secret Manager, Cloud Scheduler, Cloud Storage; Cloudflare Pages; Docker, WSL2

WORK EXPERIENCE

Poolwerx Corporation Pty Ltd
Australia (Remote)

Data and Inventory Transformation Specialist
Jun 2025 - Present

Flagship build: PWX Tool Trackers - internal franchise-operations platform, designed, built and operated end-to-end as sole developer (AI-assisted with Claude Code)

- Shipped a production web platform used daily by the support team to manage product, inventory and cost operations across 140+ franchises (AU/NZ/US): 13 operational tools and 159 API routes on Next.js 14, TypeScript, Tailwind CSS and Supabase (PostgreSQL), deployed as a Docker container on Google Cloud Run behind a custom HTTPS domain with Cloud Build deploys, GCP Secret Manager and Cloud Scheduler jobs
- **Reply Assistant** (Oct 2026): an autonomous Gemini agent that answers franchise replies on Freshdesk support tickets. A webhook plus a 15-minute poll hands each reply to the agent, which classifies it (ignore, answer from the knowledge base, investigate, or hand to a person) and investigates in a function-calling loop of up to 8 read-only tool calls over COGS lines, the store's own Lightspeed products and sales, and tracker state. Drafts are JSON against a response schema (temperature 0.1, three-model fallback chain), land as a private note and Inbox entry with 85%+ confidence marked "Ready to send", and auto-send only in live mode; approved replies become learned answers and a daily harvest of closed tickets proposes new ones
- Evaluated before trusted: a replay harness re-runs real historical tickets with Gemini as the grader, so prompt changes are measured before they ship; reordering the prompt for Gemini's implicit cache lifted cached input tokens from 10% to about 50%, bringing a ticket to about 2 US cents; every Gemini call is audit-logged with token counts behind one key and daily budget per feature
- **APL Product Mapping** for the ERP/MDM migration: deterministic tiers (learned matches, supplier part number, barcode, exact and fuzzy name, model number) map each franchise's Lightspeed catalogue to the approved product list, then Gemini returns a structured verdict with confidence on the top-10 retrieval candidates; the auto-accept threshold is set by a replay test at 97%+ precision; the first store's unmapped products fell from 2,348 to 1,450 of 3,701, and confirmed pairings carry over network-wide
- Security-first architecture: domain-restricted Google OAuth (NextAuth v5) with per-user approval and admin role-based access; default-deny Postgres row-level security with every read/write flowing through server-side API routes; per-store Lightspeed API tokens (~150 stores) held in GCP Secret Manager, never exposed to the browser or logs; every write logged with before/after JSON snapshots behind a filterable audit-log viewer with one-click undo
- Wired Jira, Freshdesk and Confluence into one dashboard (read-only REST clients, per-user scoped API tokens in Secret Manager), with a Gemini AI Summary suggesting the right knowledge-base article for each ticket and an "Ask Gemini" chat answering questions from aggregated team analytics only

- Replaced manual monthly reporting with automated Lightspeed POS API exports: a parallelised API walker generates 3 years of monthly sales, item-count and product reports in minutes (previously dozens of manual downloads per store), writing paired Excel files into an auto-organised Drive folder tree, each engine verified cell-for-cell against the legacy report; one-click stock-take exports to formatted Google Sheets
- Built store-migration tools that clone a Lightspeed account's suppliers, price books and full product catalogue (variant families, composites, multi-SKU codes) onto a new account via the API, importing the data Lightspeed's own export drops, every write shape probe-verified against the live system
- Fix product categories network-wide with a consensus vote across every franchise's catalogue and Gemini classifying the rest, validated against the real category list; update a product's price across every franchise in one run, each store re-verified live before any write, with HMAC-signed scan receipts and dry-run-by-default pushes
- Shipped a no-code tracker builder so anyone on the team can spin up shareable typed tables (metadata-driven storage, no runtime DDL) with roles and sharing, live summaries and quick filters, server-side automations and one-click archiving to Google Sheets
- **Tutorial-video kit:** turns screenshots plus a YAML storyboard into narrated, branded 1080p tutorials with subtitles - FFmpeg camera moves and compositing, every graphic drawn in HTML by headless Chromium, Google Cloud Text-to-Speech narration, a content-hash cache so re-renders rebuild only changed beats, versioned outputs and a QA contact sheet. Two franchise-facing tutorials (14 and 36 minutes, 355 narrated beats) published to the shared drive and Confluence KB; built a Cloud Run Jobs renderer and a Claude Project kit so teammates write storyboards without code

Broader role

- Support and troubleshoot the Lightspeed-Xero-Vital trading terms integration (including auth issues after Lightspeed's move to scoped application tokens), and automate Lightspeed catalogue management via the X-Series REST API with bulk category creation, deduplication and leaky-bucket rate-limit handling
- Rebuilt a 2,000-SKU product average-cost workbook with GP% calculations, validation rules and exception highlighting, and built a COGS automation tool in Google Apps Script covering product list imports, SKU sales processing, monthly posting workflows with data freezing, register-to-outlet mapping and journal exports to Xero
- Write ad hoc SQL queries in Snowflake for product, inventory and reporting analysis; act as business-process SME supporting franchise partners via Freshdesk and email

Data Migration Executive — May 2024 - Jun 2025

- Performed end-to-end data processing (cleaning, validation, transformation, extraction) for migrations across tech stacks for 140+ franchise partners, ensuring accuracy, referential integrity and system compatibility through rule-based validation and QA testing
- Built VBA macro tools that automate repetitive ETL, consolidate stakeholder reports, reconcile and deduplicate records and flag errors by rule, and Google Apps Script tools for multi-sheet processing, Sheets API integration, automated status emails, Drive file management and migration-readiness dashboards per franchise

Kreloses PLT (SaaS Platform) — Malaysia (Remote)

Data Analyst / Data Migration Specialist — Full-time Nov 2022 - May 2024; Part-time May 2024 - Jun 2025

- Restructured raw data from varied sources (free-form Excel, POS exports, SQL dumps) into import-ready formats matching the platform's schema, using C# (LINQ) tools for complex transformations and SQL (MySQL, SSMS, Firebird) to filter, clean and export
- Conducted post-migration QA against client expectations (record counts, format compliance) and liaised directly with clients to define goals and manage transitions from legacy systems

IQVIA — Philippines (Hybrid)

Data Analyst / Spreadsheet Specialist — Feb 2018 - Apr 2022

- Point of contact for the Product Reference Management (Middle East and Africa) team, leading weekly and monthly production meetings and coordinating with global teams
- Self-taught Excel VBA and built automation tools for QC, data input and product processing (an automated QC tool with role-based templates, a consumer health product tracker, a patent intelligence tool consolidating 60+ global pharmaceutical datasets); received the IQVIA Impact Program Award for innovations improving data efficiency and quality control

Optum — Philippines (Onsite)

Pharmacy Benefits Analyst — Aug 2017 - Jan 2018

- Reviewed pharmacy benefit requests and gathered required data for accurate, timely implementation for US healthcare clients in a high-volume, deadline-driven environment

PERSONAL PROJECT

SukliCo POS — Oct 2026 - present

Offline-first mobile point of sale for sari-sari and small grocery stores in the Philippines, built as a multi-tenant subscription SaaS. Sole developer, AI-assisted. Launch-ready; closed testing on Google Play is the next step. Private repository, code available on request.

- Offline-first by design: a sale completes with zero internet. Writes land in SQLite and an outbox syncs them with client-generated UUIDs and idempotent upserts, so retries never conflict and the server never refuses an offline sale
- Supabase backend with multi-tenant row-level security from day one and RPC-only writes (the app holds only the anon key); an append-only ledger in integer centavos with weighted average cost and FEFO expiry batches; offline staff PIN switching with a device-keyed verifier; owner-toggled permissions enforced server-side with a full audit log; a TOTP-gated operator role
- Expo SDK 57 / React Native app with Google Play Billing (server-side verification, real-time developer notifications), a Kotlin notification-listener module for e-wallet payment confirmation, Bluetooth 58 mm receipt printing, barcode and QR scanning, and an owner web console on Cloudflare Pages; checkout, split payments, customer credit ("utang"), loyalty, shifts, stock with expiry, profit reports and an online catalogue
- Launch-ready in 18 days of commits, backed by about 385 Jest tests against real SQLite, about 1,000 pgTAP assertions on tenant isolation, roles, the ledger and billing, 63 migrations, and pre-launch security and architecture reviews; marketing explainer built in Remotion with Gemini-generated scenes and a Gemini-TTS Taglish voiceover

EDUCATION & LANGUAGES

Bachelor of Science in Pharmacy, Saint Louis University, 2016 | English (Fluent), Tagalog (Fluent)